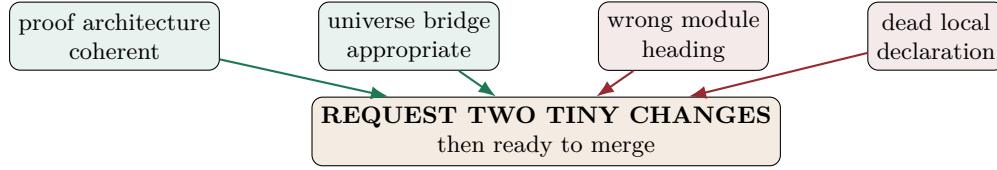


Final readiness review: InjectiveDimension

Lean 4 practice · categorical shape · merge decision



Decision

As posted	Not quite ready: two source cleanups should land first.
After fixes	Ready from the proof/API review perspective.
Validation	Run the target Mathlib build, linter, and minimum-import check on the PR branch.

Required before merge

1. Correct the file heading.

```
- # Projective Dimension in ModuleCat
+ # Injective Dimension in ModuleCat
```

A one-line scope sentence should mention invariance under linear and ring-semilinear equivalences, including different universes.

2. Delete the unused declaration; use **exact**.

```
- have := exactS.hasInjectiveDimensionLT_X3_iff n inferInstance
- apply (exactS.hasInjectiveDimensionLT_X3_iff n inferInstance).symm.trans
+ exact (exactS.hasInjectiveDimensionLT_X3_iff n inferInstance).symm.trans
  ((ih eCoker).trans
    (exactS'.hasInjectiveDimensionLT_X3_iff n inferInstance))
```

The original anonymous **have** is immediately recomputed and never consumed. Naming the quotient equivalence **eCoker** also avoids overloading **e**.

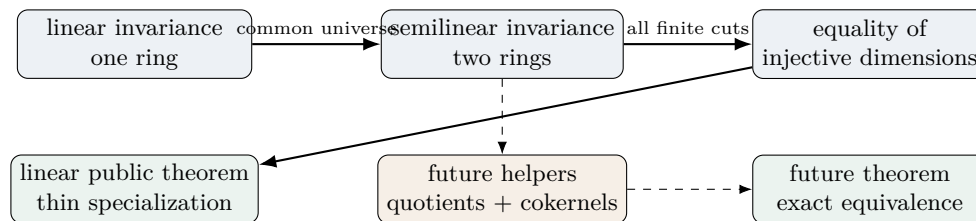
Proof shape: keep

$$\begin{array}{ccccccc}
 0 & \longrightarrow & M & \xrightarrow{f} & I & \xrightarrow{q} & C = I / \text{range}(f) \longrightarrow 0 \\
 & & \downarrow e_{MN} \sim & & \downarrow e_I \sim & & \downarrow e_C \sim \\
 0 & \longrightarrow & N & \xrightarrow{f'} & I' & \xrightarrow{q'} & C' = I' / \text{range}(f') \longrightarrow 0
 \end{array}$$

$$\text{idim}(M) \leq n + 1 \iff \text{idim}(C) \leq n \iff \text{idim}(C') \leq n \iff \text{idim}(N) \leq n + 1.$$

This is the correct induction invariant: injective middle terms, equivalent cokernels, and dimension shifting. The base case correctly reduces dimension zero to injectivity and transports it through Baer's criterion / ring equivalence.

Lean and category-theory review



Good practice to preserve

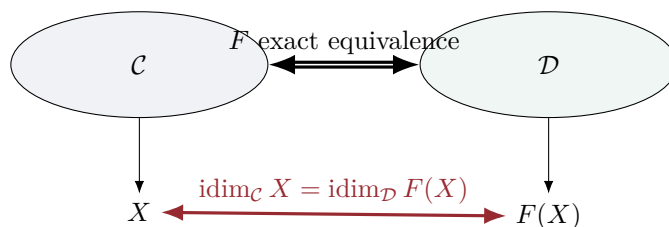
- The public linear result is a one-line specialization of the semilinear theorem.
- `ULift` moves both modules explicitly to $\max(v, v')$; this is clearer than hiding the universe transport behind coercions.
- `RingHomInvPair.of_ringEquiv` and the transparency option have narrow lexical scope.
- The new root import is alphabetically placed after `Injective`.
- The theorem states the strongest useful cross-ring, cross-universe result.

Small source-quality improvements

- Put `hasInjectiveDimensionLE_iff_of_linearEquiv_aux` in a section that does not declare unused `R'` and outer `e`; use `eMN`, `eI`, `eCoker`.
- Rename the extended-natural cutoff in `eq_of_forall_ge_iff` from `N` to `d`; `N` already denotes the target module.
- Name injectivity evidence (for example `injI'` and `injI`) rather than relying on anonymous local instances when practical.

These improve readability but need not expand this PR beyond the two required edits.

Follow-up API, not a blocker



Prove injective-dimension invariance once for exact equivalences of abelian categories. Separately, add a constructor transporting a semilinear equivalence to quotients when the relevant submodules map to one another, and package the canonical short exact sequence of an injective linear map. Those abstractions would remove the low-level `mapQ/range` bookkeeping from a later refactor.

Final checklist

☐ heading corrected	☐ dead have removed
☐ target file builds	☐ linter / minimum imports pass
☐ no sorry or new axioms	☐ generated root import checked

Bottom line: the mathematics and induction design look mergeable. Apply the title and dead-code fixes, verify on the actual PR branch, and approve.